

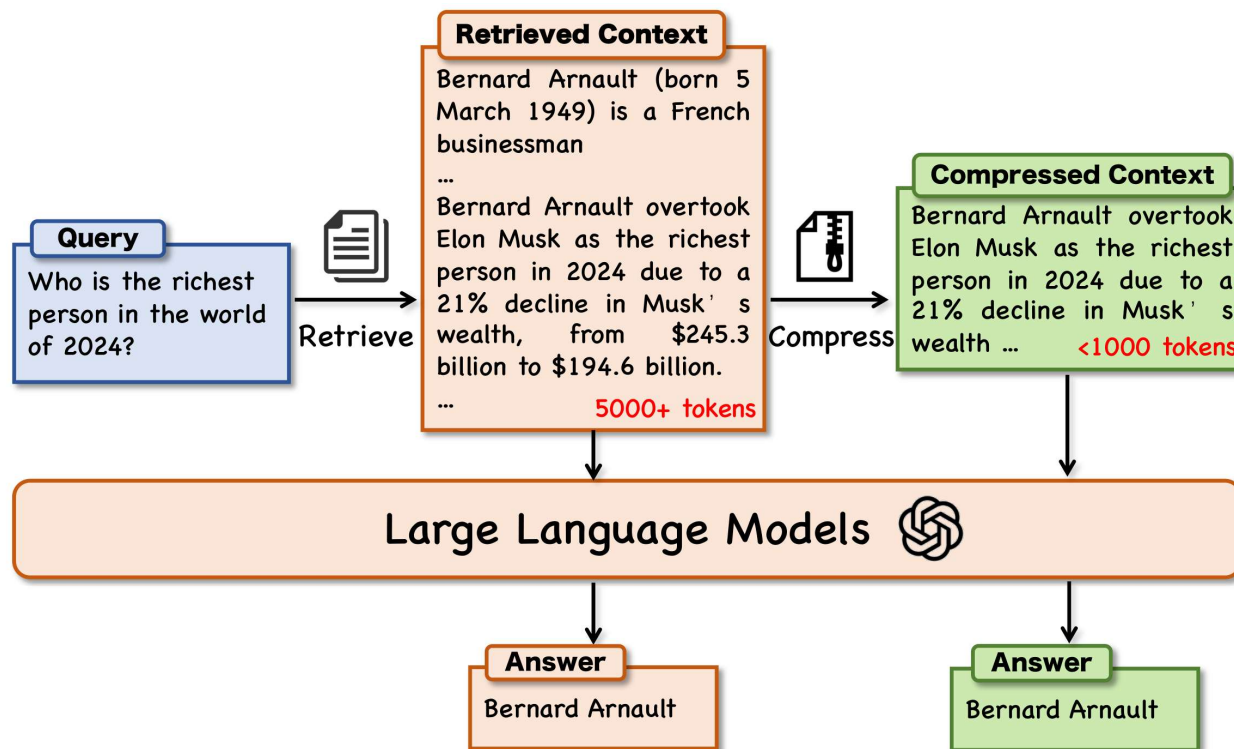
AttentionRAG: Attention-Guided Context Pruning in Retrieval-Augmented Generation

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Background: Challenges

- The long nature of RAG contexts exacerbates **attention dilution**
- The sentence-based nature of the query makes it challenging to directly **align** with critical content in the context.



Related Work: LLMingua

- LLMingua compresses prompts by using a **budget controller** that allocates varying compression ratios to different components of the prompt.
- **Lacking context-awareness** makes it challenging to determine the **optimal compression ratio** for a given LLM.

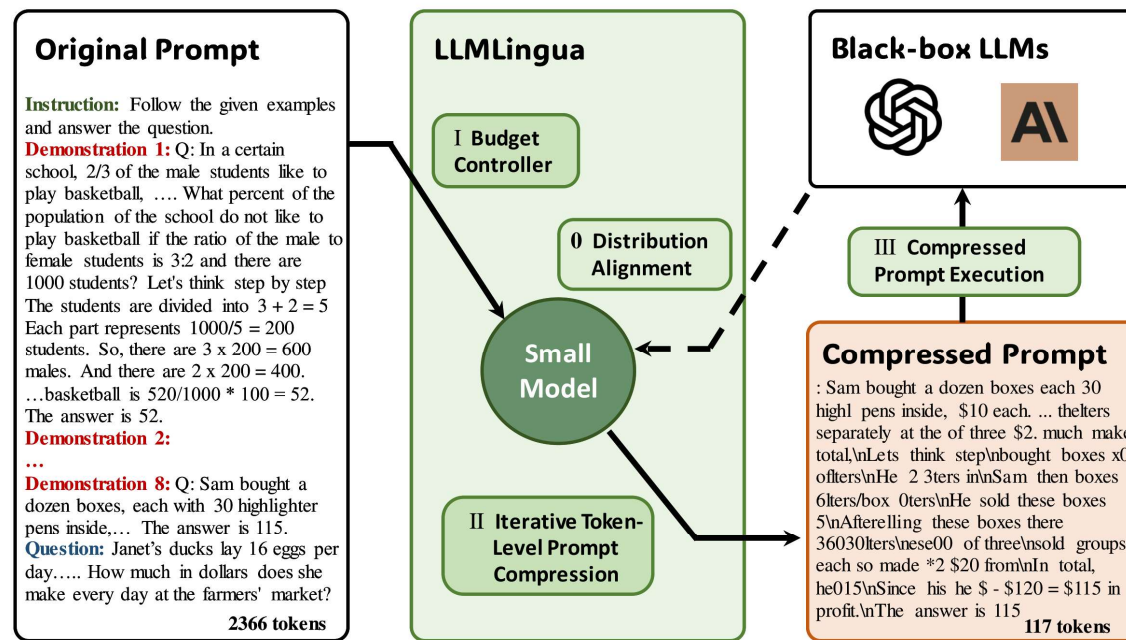


Figure 1: Framework of the proposed approach LLMingua.

AttentionRAG: Overall Methods

- Construct Answer Hint **Prefix**
- Compute **Attention Features** from LLM
- **Compress** with Attention

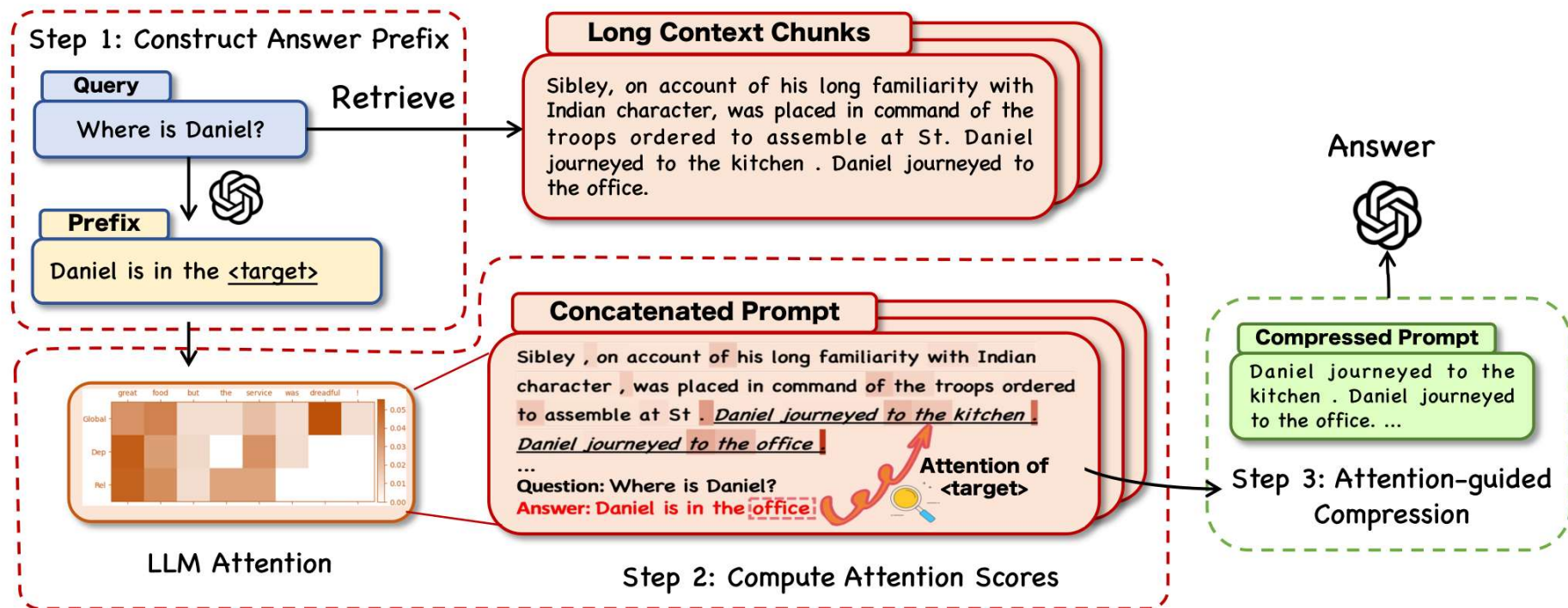
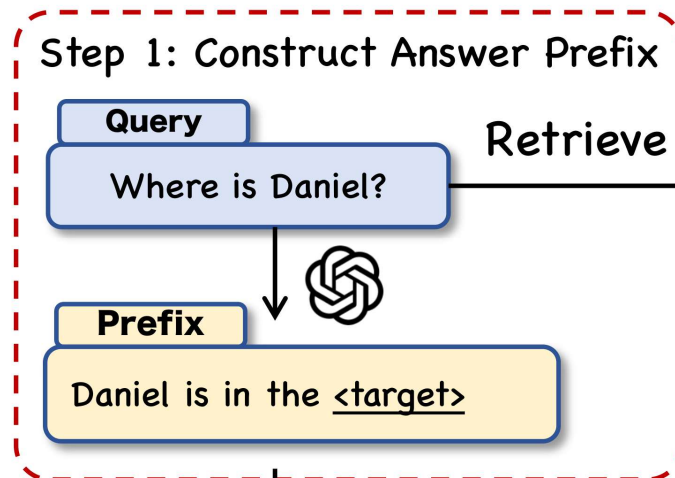


Figure 2: Illustration of ATTENTIONRAG

AttentionRAG: Construct Answer Hint Prefix

- Associate each query with an **answer hint prefix**
 - allows the LLM to **calculate the query-context attention** through one focal token.
 - improve the **alignment** between the query and context
- Categorize the answer hint prefix into two types
 - Empty Yes/No questions
 - Non-empty Wh-questions
- Purpose: focus attention on the focal token



You are a formatting assistant. Given a question, your task is to generate a corresponding answering format. The format should maintain the same structure as the question but transform it into an incomplete answer template. If it is impossible to generate a format, return “None”.

The format is like an complete answer, but truncated before the key word, and the key word is not included in the format.

For instance, if the question is “Where is Daniel?”, the format should be “Daniel is in the”, as the next word is the key word.

Note: For yes/no questions, such as “Is Tom here?”, return “None” because these questions are typically answered with “yes” or “no” and do not have a natural continuation that leads to a single keyword.

AttentionRAG: Compute Attention Features from LLM

- **Divide** the retrieved context C produced by the RAG framework into smaller chunks $c_1 \dots c_n$.
- **Concatenate** the chunked context, query, instruction and the generated answer hint prefix and feed this into the LLM.
- **Attention** features:

$$A_j = \sum_{l=0}^L \text{Attention}_l(c_j, a_j)$$

- Capture both local and global dependencies in the input.

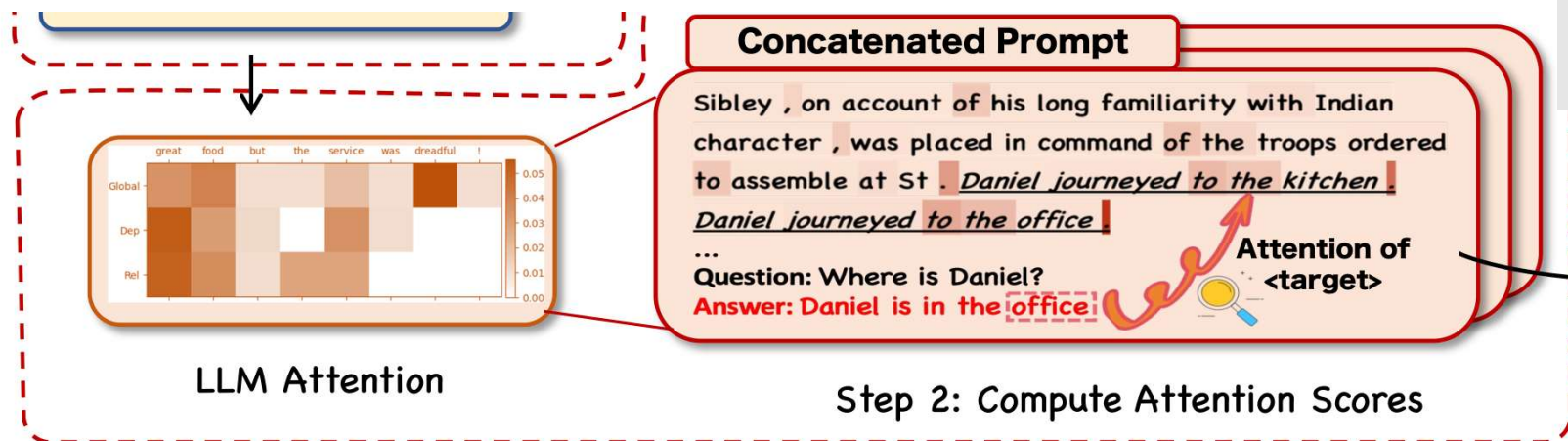
You will be given a long context begin with 'Context:', a question begin with 'Question:', and a hint begin with 'Hint:'. Please answer the question.

Context: {chunk}

Hint: You should answer begin with {prefix_hint}, if there is no useful information in the context for the question in the context and you really don't know the answer, just answer prefix_hint none.

Question: {question}

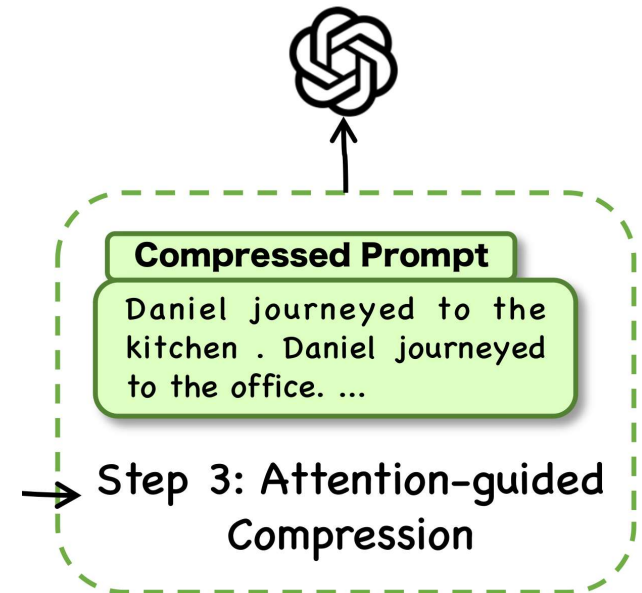
Answer:
{prefix_hint}



AttentionRAG: Compress with Attention

- check whether this token is "none. "
 - Skip
- identifying the tokens in the context that have the highest attention features with respect to the focal token
- select the top-k tokens.
 - focus on the sentences that contain these top-k tokens
 - concatenated to form a compressed context

$$c'_j = \text{Concat}(\{s \mid t_r \in \text{Top-}k(A_j) \text{ and } t_r \in s\})$$



AttentionRAG: Pseudocode

Algorithm 1 Pseudocode of ATTENTIONRAG.

```
1: Input: Retrieved long context  $C$ , query  $q$ , generation model  $L$ , compression model  $L_C$ 
2: Output: Generated sequence  $y$ 
3:
4: Generate Answer Hint Prefix
5: Get answer hint prefix  $p$  through  $L$  with  $q$ 
6: Chunking
7: Generate chunks  $c_1, \dots, c_n$  by partitioning  $C$  with chunk size  $m$ , where  $n = \lceil |C|/m \rceil$ 
8: Initialize empty variable  $C'$ 
```

```
9: Compressing with Attention
10: for  $j = 1$  to  $n$  do
11:   Generate the anchor token  $a_1$  with  $L_C$ ,  $c_j$ ,  $q$ , and  $p$ 
12:   if  $a_1$  is "none" then
13:     continue
14:   else
15:     Obtain Attention Features  $A_1$  with  $a_1$  and  $c_j$  ▷ Eq. (2)
16:     Get compressed  $c'_j$  according to  $A_1$  and  $c_j$  ▷ Eq. (3)
17:     Append  $c'_j$  to  $C'$ 
18:   end if
19: end for
20: Generate  $y$  from  $L$  with  $C'$  and  $q$ 
21: Return Generated sequence  $y$ 
```

Experimental Setup

- **Models:** Llama-3.1-8B-Instruct, Qwen-2.5-7B-Instruct
- **Baselines:** LLMingua2, LongLLMingua
- **Datasets:** LongBench, BABILong
- **Metrics:**
 - Exact Match (EM)
 - LLM-as-a-judge scores
 - Compared with GPT-4
 - Compression Ratio (CR)

Dataset	Chunk_Size	K
HotpotQA	300	12
2WikiMQA	300	15
TriviaQA	150	8
BABILong 1k	50	8
BABILong 2k	100	10
BABILong 4k	200	12

Table 6: Hyperparameter settings of the experiment

$$CR = \frac{\# \text{ tokens in original context}}{\# \text{ tokens in compressed context}}$$

Results: Performance in LongBench

Model	Method	2WikiMQA			HotpotQA			TriviaQA		
		EM↑	LLM Judge↑	CR↑	EM↑	LLM Judge↑	CR↑	EM↑	LLM Judge↑	CR↑
Llama-3.1-8B-Instruct	Original Prompt	0.49	0.46	1.0x	0.50	0.62	1.0x	0.87	0.80	1.0x
	Random	0.21	0.15	5.0x	0.05	0.01	3.3x	0.76	0.73	1.7x
	LLMLingua2 (small)	0.24	0.26	5.0x	0.40	0.45	3.3x	0.86	0.82	1.7x
	LLMLingua2 (large)	0.33	0.30	5.0x	0.44	0.51	3.3x	0.84	0.76	1.7x
	LongLLMLingua	0.23	0.22	3.3x	0.33	0.40	3.3x	0.83	0.72	1.7x
	ATTENTIONRAG (small)	0.39	0.36	6.3x	0.45	0.55	3.7x	0.84	0.77	2.0x
	ATTENTIONRAG (large)	0.42	0.38	15x	0.48	0.61	5.6x	0.89	<u>0.81</u>	1.7x
Qwen2.5-7B-Instruct	Original Prompt	0.56	0.46	1.0x	0.58	0.68	1.0x	0.94	0.82	1.0x
	Random	0.18	0.06	5.0x	0.04	0.01	3.3x	0.65	0.51	1.7x
	LLMLingua2 (small)	0.29	0.27	5.0x	0.43	0.47	3.3x	0.91	0.80	1.7x
	LLMLingua2 (large)	0.40	0.28	5.0x	0.48	0.58	3.3x	0.87	0.81	1.7x
	LongLLMLingua	0.31	0.18	4.0x	0.28	0.31	3.3x	0.83	0.73	1.7x
	ATTENTIONRAG (small)	0.41	0.28	6.3x	0.51	0.54	3.7x	0.83	0.73	2.0x
	ATTENTIONRAG (large)	0.41	0.30	15x	0.53	<u>0.56</u>	5.6x	<u>0.88</u>	<u>0.80</u>	1.7x

Table 1: Performance of different methods in 2WikiMQA, HotpotQA and TriviaQA dataset.

Results: Performance in BABILong

Model	Method	1K			2K			4K		
		EM↑	LLM Judge↑	CR↑	EM↑	LLM Judge↑	CR↑	EM↑	LLM Judge↑	CR↑
Llama-3.1-8B-Instruct	Original Prompt	0.74	0.64	1.0x	0.67	0.57	1.0x	0.71	0.60	1.0x
	Random	0.11	0.04	2.9x	0.09	0.05	3.3x	0.05	0.04	3.3x
	LLMLingua2 (small)	0.55	0.41	2.9x	0.38	0.29	3.3x	0.36	0.29	3.3x
	LLMLingua2 (large)	0.69	0.51	2.9x	0.53	0.41	3.3x	0.49	0.37	3.3x
	LongLLMLingua	0.63	0.59	2.9x	0.43	0.38	3.3x	0.47	0.36	3.3x
	ATTENTIONRAG (8B)	0.74	0.65	3.8x	0.58	0.49	3.8x	0.53	0.41	5.6x
Qwen2.5-7B-Instruct	Original Prompt	0.87	0.78	1.0x	0.75	0.71	1.0x	0.80	0.75	1.0x
	Random	0.04	0.04	2.9x	0.05	0.03	3.3x	0.04	0.01	3.3x
	LLMLingua2 (small)	0.49	0.38	2.9x	0.37	0.30	3.3x	0.34	0.24	3.3x
	LLMLingua2 (large)	0.70	0.52	2.9x	0.55	0.41	3.3x	0.50	0.38	3.3x
	LongLLMLingua	0.54	0.47	2.9x	0.35	0.29	3.3x	0.37	0.29	3.3x
	ATTENTIONRAG (8B)	0.88	0.76	3.8x	0.62	0.57	3.8x	0.66	0.59	5.6x

Table 2: Comparison of different methods on BABILong.

Enhance overall response quality, effectively balancing compression and clarity.

Query:	Where was the wife of Francis I Rákóczi born?	Answer: Ozalj
Original Context:	Passage 1: Waldrada of Lotharingia Waldrada was the mistress, and later the wife, of Lothair II of Lotharingia. Biography Waldrada’s family origin is uncertain. The prolific 19th-century French writer Baron Ernouf suggested that Waldrada was of noble Gallo-Roman descent, sister of Thietgaud... (7003 tokens)	city of Gyulafehérvár, Transylvania. ✗
Random:	drada,, of. th-century French Baron Er of-R sister ofga bishopther arch of, not any socialoli,... (1400 tokens)	The text does mention it. ✗
LLMLingua2:	Waldrada Lotharingia mistress Lothair II Gallo sister Thietgaud niece Gunther Vita Sancti related Eberhard II Etichonids 855 Lothar II married Teutberga 858 862 Nicholas 863Charles ...(632 tokens)	Munkács.✗
LongLLMLingua:	Passage:Waldrada theressairia. is The proific 1th French Baron Ern thatadaoman, sister of Th Trier, Gun of and have suggested of social though anatic.itactoli thatada Ehard II,edbourgichon (920 tokens)	Hungary. ✗
ATTENTIONRAG:	... Life Early years and family Ilona was born Ilona Zrínyi in Ozalj ... She was the daughter of Petar Zrinski, Ban (viceroy) of Croatia, the niece of both Miklós Zrínyi and Fran Krsto Frankopan and the wife of Francis Rákóczi I ... (273 tokens)	Ozalj ✓

Table 3: Examples of compression results by various methods

Results: Ablation Study

- Efficiency Analysis
- Attention Layer Choice
- Hyperparameters
 - introduce more dynamic
- Size of Compression Models
 - smaller models can achieve similar results

Layer Subsets	EM↑	LLM Judge ↑	CR↑
0 - 10	0.35	0.43	4.5x
11 - 20	0.38	0.50	3.6x
21 - 31	0.40	0.48	3.7x
0 - 31	0.42	0.54	3.6x

Table 4: Performance on HotpotQA with different subset of layers

Chunk Size	Top-K	EM↑	LLM Judge ↑	CR↑
300	5	0.80	0.73	3.2x
300	10	0.84	0.75	2.2x
300	15	0.87	0.77	1.9x
100	10	0.90	0.79	1.9x
200	10	0.88	0.77	2.2x
400	10	0.85	0.77	2.4x

Table 5: Performance on TriviaQA with different chunk sizes and top-K values

Limitations

- **Complex queries** cannot fully address
- **Computing attention features** need to be optimized using more sophisticated algorithms to improve **efficiency**
- not developed **explicit methods** for controlling or instructing the desired ratio

Summary

Problem

- RAG systems face challenges with long retrieved contexts
- Existing methods lack context awareness
- dilution and alignment issues.

Solution

- Construct Answer Hint Prefix
- Compute Attention Features from LLM
- Compress with Attention

Result

- Achieve high compression ratios and good performance.
- Compress context effectively with better - quality.